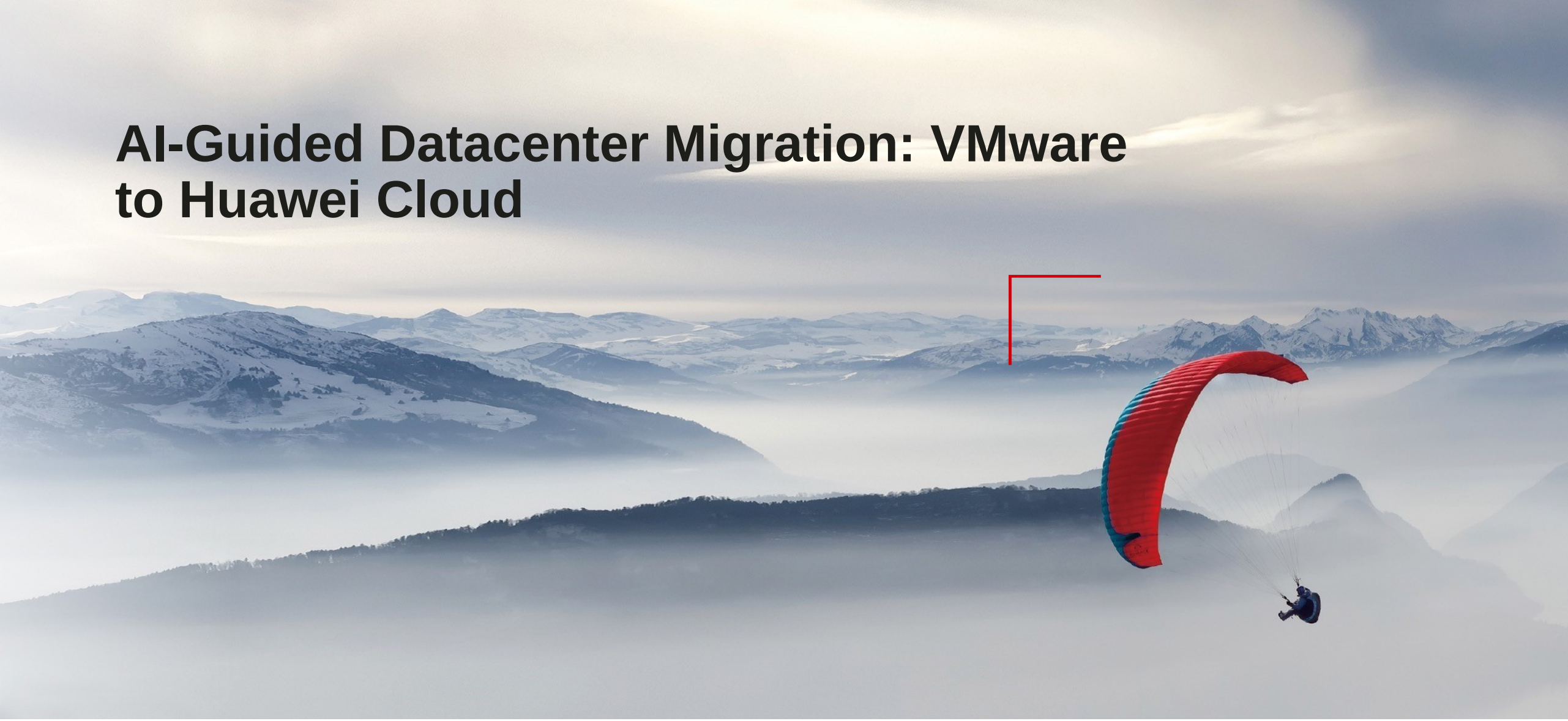
AI-Guided Datacenter Migration: VMware to Huawei Cloud



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AI-Assisted Migration: Usage Guidelines

Practical guidance, not a formal delivery commitment

How to work with AI for VM-to-ECS migration

How to Use This Guide

This guide shows what to ask your AI assistant, what knowledge it needs, and how to verify its work to prevent errors. AI accelerates migration planning and execution -- but does not replace expert review or customer decisions.

1. What AI Does

Planning / Analysis / Automation

- AI plans, analyzes, generates scripts, and assesses risks
- It does not replace design, expert review, testing, or decisions
- Review all outputs before acting on them

2. What to Provide

Anonymized Data Only

- Use simulated, test, or anonymized data only
- Never enter real credentials, passwords, keys, or customer data
- Respect security, compliance, and confidentiality

3. How to Work with AI

Review / Authorize / Verify

- Review every AI output before acting
- Authorize each migration step explicitly
- Keep a human in the loop for all production changes

4. When to Be Cautious

Sensitive Systems / Manual Review

- For sensitive systems (financial, defense, classified), consider manual migration
- Use strict AI oversight for compliance-heavy workloads
- Always prioritize security and sovereignty requirements

AI-assisted migration improves efficiency; it does not replace human judgment. Verify every step against real data.

Your VMware Migration Scenario

Your Starting Point

You have a VMware environment with 100+ virtual machines running your business applications. Licensing costs are rising. You need a plan to move to the cloud.

Your Goal

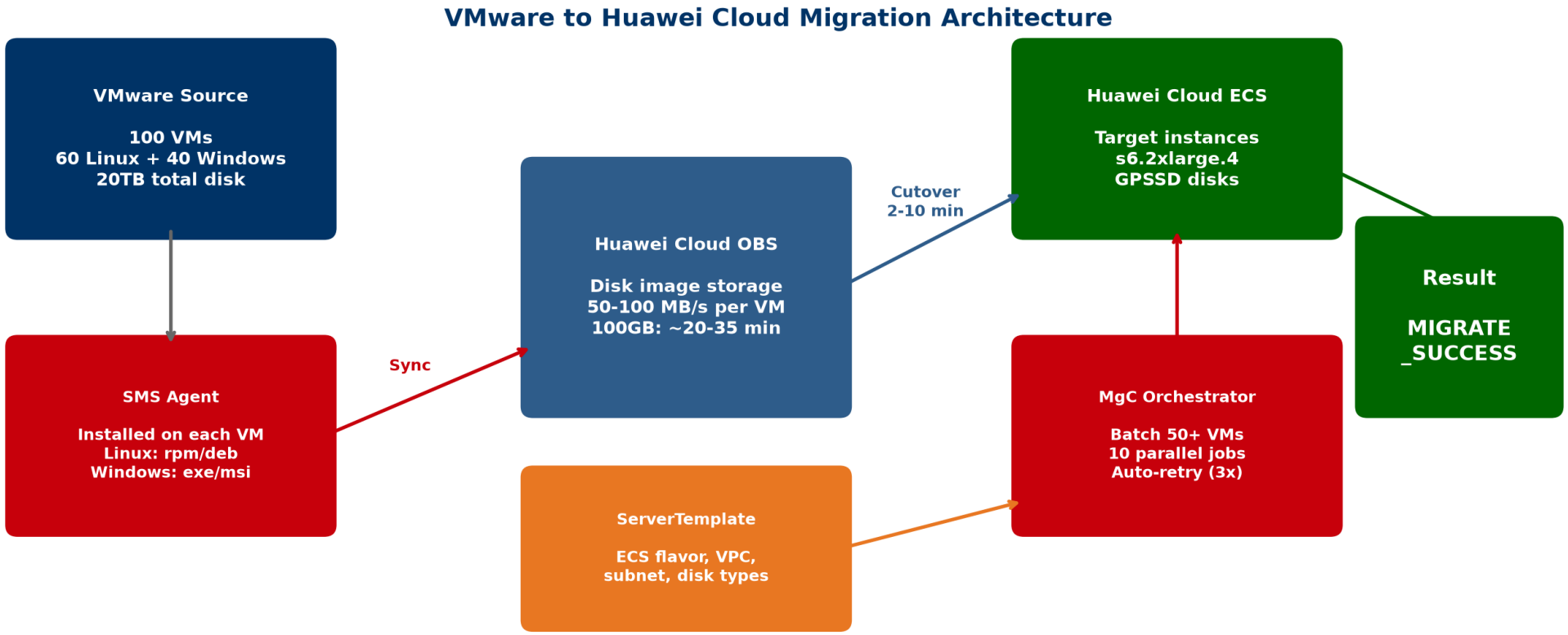
Move everything to Huawei Cloud without downtime. Keep your applications running. Reduce costs. Modernize your infrastructure.

Why AI-Assisted

AI can scan your entire environment in minutes, map all dependencies automatically, and generate a step-by-step migration plan. What used to take weeks of manual planning now takes days.

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Step 1: Ask AI to Discover Your Environment



Example prompt: "Review my VMware environment and list all virtual machines, their operating systems, storage, and how they connect to each other"

AI Should Know: How to read your vCenter inventory, identify which VMs depend on each other, and group them by application

Source: Huawei Cloud SMS Documentation (https://support.huaweicloud.com/sms/) — MgC Edge Discovery Guide, Accessed: Aug 2026.

Verify: Compare the AI's list against your actual environment. Make sure no VMs are missing and all dependencies are captured

Prevent Hallucinations: The AI should reference real VM names from your environment, not make them up

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Step 2: Ask AI to Prepare Migration Templates

What to Ask AI

Example prompt: "Prepare a migration plan for each group of VMs. Choose target server sizes that match or exceed what we have today. Set up networking and security for the new environment."



AI Should Know & Verify

The available server sizes on Huawei Cloud, how to map your current storage to cloud storage, and how to configure networks securely.
Verify: Each planned server is properly sized -- not too small, not wastefully large. AI should use real options from the catalog, not guesses.



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Step 3: Ask AI to Execute Migration

Migration Job Execution: Two-Phase Process

Phase 1: Background Sync

- Agent uploads disk data to OBS
- Non-disruptive — VM stays running
- Sync speed: 50-100 MB/s per VM
- 100GB disk: ~20-35 min
- Monitor: sync %, data transferred, ETA

Ready →

Phase 2: Cutover

- Brief downtime: 2-10 min per VM
- Final incremental sync (<5% of disk)
- ECS creation: ~30 seconds
- Switch traffic to new ECS
- Decommission source after sign-off



On failure: review logs → fix issue → retry from last checkpoint

Source: Huawei Cloud SMS Migration Workflow (https://support.huaweicloud.com/sms/) — 7-Phase Process. Accessed: Aug 2026.

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Step 4: Ask AI to Scale and Handle Errors

What to Ask AI

Example prompt: "Scale up and migrate the remaining VMs in batches, grouped by application. Start with the least critical ones first. Handle any errors automatically."



AI Should Know

Common migration issues and how to fix them. When a VM won't migrate one way, try an alternative approach. How to pause and resume if needed.



Verify & Prevent Hallucinations

When the AI says an error is 'fixed', verify it yourself. The AI should explain what went wrong and what it did, not just say 'resolved'. Never assume a retry succeeded without checking.



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Step 5: Ask AI to Validate Every VM

What to Ask AI

Example prompt: "Check every migrated VM -- does it boot? Is the network working? Are all services running? Give me a report with specific results for each VM."



AI Should Know

What a successful migration looks like, how to compare data before and after, and what performance metrics matter for your applications.



Verify & Prevent Hallucinations

Don't accept 'everything looks good'. Ask for proof: boot logs, network tests, data comparisons. Compare performance with your original VMs. A human must approve before going live.



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How to Make Your AI a Migration Expert

Teach It the Process

Tell your AI what success looks like: which VMs to migrate, acceptable downtime, and validation criteria. Clear instructions produce better results.



Give It Real Data

Connect your AI to your Huawei Cloud account and documentation. It should know your actual server sizes, networks, and storage options -- not guess or assume.



Set Clear Expectations

Install migration skills from our GitHub repository. Point your AI to Huawei Cloud documentation. Works with OpenCode, Hermes, or Claude Code.



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Why AI-Assisted Migration

Speed & Efficiency

Planning that used to take weeks now takes days. AI discovers 1000+ servers in 10 minutes. 40% faster migration with automated orchestration.



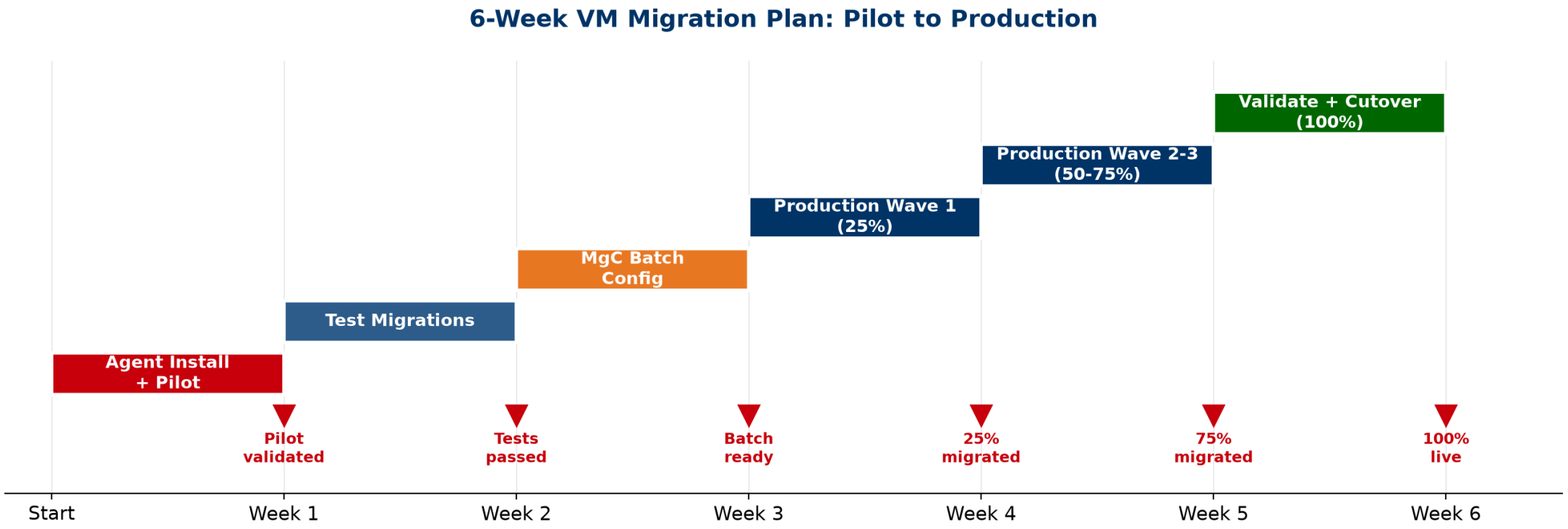
Risk Reduction

AI spots incompatibilities and dependencies early. If a VM won't migrate one way, it automatically tries another. Over 95% succeed on the first try.



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Your 6-Week AI-Guided Journey



Week 1 -- Example prompt: "Set up and test migration with 2-3 non-critical VMs. Confirm everything works."
Week 2 -- Example prompt: "Test the full migration process end-to-end. Fix any issues."
Week 3-5 -- Example prompt: "Migrate in waves -- 25% at a time, by application. Verify each wave before proceeding."
Week 6 -- Example prompt: "Run final checks. Switch DNS to the new environment. Retire old VMs."

Source: Huawei Cloud Migration Best Practices (https://support.huaweicloud.com/sms/) — 6-Week Migration Plan. Accessed: Aug 2026.

Equip Your AI Assistant

Install Migration Skills:
https://github.com/binrogithub/1-3-Cloud-Adoption-Skills/tree/main/AI-Assist-Migration/Datacenter-on-premise-migration

AI Tools: OpenCode, Hermes, or Claude Code

Getting Started:
1. Install skills in ~/.opencode/skills/
2. Open your AI coding tool
3. Example prompt: "Help me migrate my VMware estate to Huawei Cloud"

Skills include: step-by-step guides, automation scripts, and runbooks

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Thank you.

Bring digital to every person, home and organization for a fully connected, intelligent world.

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